

Assignment 11 – Data-Driven Automation in Python Behave Framework

1. Module Name

Python BDD Restful Automations

2. Experiment Title

Scenario 2: Use of Test Data Driven Automation in Python Behave Framework

3. Problem Statement

Develop a data-driven web automation scenario using Python, Selenium and Behave BDD so that the same login workflow can be executed with multiple sets of valid and invalid test data.

4. Objective

Implement data-driven automation using Behave Scenario Outline and Examples, execute multiple login combinations, and verify both successful and unsuccessful authentication results.

5. Tools, Software and Concepts Used

Python; Selenium WebDriver; Behave BDD; Google Chrome; Gherkin; Scenario Outline; Examples table; positive and negative test data.

6. Source Code / Implementation Steps

Created a Behave feature using Scenario Outline. Defined multiple username, password and expected-result combinations in the Examples table. Implemented reusable Python step definitions for opening the application, entering credentials, clicking Login and validating the expected result. Added environment.py for browser setup and teardown. Test data is also documented in Test_Data/login_test_data.csv.

7. Output / Labelled Screenshots

Screenshot 1 – TC01 Successful Login. Screenshot 2 – TC02 Invalid Login. Screenshot 3 – TC03 Invalid Password. Screenshot 4 – TC04 Invalid Username. Screenshot 5 – Behave Execution showing all scenarios completed. Screenshots are stored in Source_Code/Screenshots.

8. Result

The data-driven Behave automation was implemented successfully. Multiple login test-data combinations can be executed through one Scenario Outline, reducing duplicate test steps.

9. Observation

The Scenario Outline and Examples table make it possible to reuse the same automation flow for multiple data sets. Both positive and negative login cases can be maintained in a structured and readable format.

10. Conclusion

The experiment successfully demonstrated data-driven automation using Python, Selenium and Behave BDD. The approach provides a scalable foundation for adding further test-data combinations and business scenarios.